

GeoSpatial Data Management 2025

Final project

We present two alternative projects. The students of the GDM course have to select one of these. The instructions for the delivery and presentation of the project can be found at the bottom of this page

Project 1: transportation means analysis

The goal is to collect a personal trajectory and analyze it using the methods presented during the course or similar. Specifically, the trajectory should report the movement conducted using different transportation means, for example walking and train. The student shall record WHEN the different transportation means are used. For example, assume to experience: walking – train –walking: for each of the three phases, record the time intervals along with the transportation mean used. This description represents the GROUND TRUTH.

1. Choose a location tracking app (example: Geo tracker, Strava). The only requirement is the capability of exporting data in GPX format
2. Collect the trajectory:
 - a. Plan data collection in advance
 - b. Collect the trajectory along the ground truth using the app
 - c. Import data
3. Display statistics on the data
4. (optional) Pre-process the data, if needed
5. Display the speed variation in the trajectory
6. Try to determine the transportation means being used. Compare the outcome with the ground truth:
 - a. Is this operation straightforward or do you see any problem?
 - b. In the latter case, can you envisage a method to solve the problem?

Project 2: Eurostat data analysis

Eurostat is the statistical office of the European Union (EU) and its mission is to provide high-quality statistics and data on Europe. For this exercise we consider the following datasets provided by GISCO, the Geographic Information System of the European Commission (<https://ec.europa.eu/eurostat/web/gisco/overview>). Note that the amount of data is significant. Therefore the suggestion is to work on a limited subset of data.

1. EU Population (Version Census 2021 (22 January 2025)), in vector format (geopackage format): <https://ec.europa.eu/eurostat/web/gisco/geodata/population-distribution/geostat>. The dataset contains various information

2. Geographic accessibility to healthcare services in vector format:
<https://ec.europa.eu/eurostat/web/gisco/geodata/geographic-accessibility#Accessibility%20to%20healthcare%20services>
3. EU communes (i.e. municipalities) in vector format:
<https://ec.europa.eu/eurostat/web/gisco/geodata/administrative-units/communes>

These datasets have the following characteristics:

- dataset 1 and dataset 2 consists of grids of squared cells of side 1km. Each cell has a position and an ID that have been defined by UE., thus the geometry of two grids is identical. Instead Dataset 3 consists of irregular polygons.
- dataset 1 contains various information on pop, including total population, male, female and so on. It does not contain any information on the municipality it belongs to
- dataset 2 contains for each cell the time (in minutes) estimated to reach by car the closest hospital

Task: select a subset of communes in Italy (e.g. municipalities in Lombardy or a subset of them). The exercise consists of computing the following information.

Q1: the amount of population by municipality. How is the population distributed?

Q2: determine the mean accessibility value to healthcare services in each municipality of the reference area

Q3. determine the total amount of population living in the municipalities where the mean accessibility value is low

Q4: compare the mean accessibility value in 2020 vs 2023 (consider the value indicating the time to reach the closest healthcare service) to determine whether there has been a progress or not

For each question, explain how it has been addressed and provide one or more maps to visually illustrate the result

Instructions

The project can be developed by single students or groups of two students.

What has to be provided:

- A report describing the work done on the chosen scenario. Specify the goal, the methodology (what has been done) and the results. The report shall to be sent at least 2 days before the exam (maria.damiani@unimi.it, fatme.hachem@unimi.it)
- A short slide presentation of 10'. The presentation will be followed by a discussion of the project